Developing Guidance for Landscape Connectivity in Forest Landscape Planning: Case Study for the Quesnel FLP

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Abstract

Landscape connectivity provides the pulse by which we can monitor the health and resilience of ecosystems. Here we demonstrate the status of landscape connectivity management in B.C. and provide a science-based pathway to implementing robust strategies. Our work is divided into four phases.

Phase 1: we review the current management strategies for landscape connectivity corridors, and ecosystem networks.

Phase 2: we conduct a systematic review of the literature to determine practical approaches to implementing landscape connectivity modelling

Phase 3: we implement Omniscape to demonstrate how to model for species-agnostic e.g., biodiversity focused, connectivity for the Quesnel Natural Resource District (DQU).

Phase 4: we take our learnings from Phase 1-3 and develop a Values, Objectives, Indicators, and Targets (VOIT) worksheet for a landscape connectivity management strategy.

In Phase 1 we found that landscape connectivity is poorly implemented as a landscape management strategy in BC, largely due to the disconnect between objective setting and analytical assessments. In Phase 2, we identified Omniscape, a tool that implements Circuitscape to perform omni-directional connectivity modelling as a practical yet highly recommended tool in scientific circles. In Phase 3, we conduct landscape connectivity modelling for biodiversity in the DQU and demonstrate the usefulness of Omniscape outputs to (1) model current connectivity, (2) develop management strategies to recover, maintain, or enhance connectivity, (3) provide a platform for routine monitoring, and (4) connectivity characterization that translates to written strategies and objectives. In Phase 4, we showcase how a potential FLP strategy for landscape connectivity may be developed.

1. Introduction

A connected landscape is a functional landscape. Connectivity across forested landscape scales i.e., within, between, and among patches facilitate the movement of ecological information that supports population viability (Kool et al., 2013), community dynamics (Damschen & Brudvig, 2012), and ecosystem recovery (Pearson et al., 2021). These ecological elements are key ingredients for the Biodiversity and Ecosystem Health Framework. However, practices and legal targets for connectivity management in B.C. remain largely under implemented due to the lack of local guidance and standardized workflows to inform verifiable and measurable objectives. On the heels of Forest Landscape Planning initiatives, the appetite for connectivity objectives underpins key FRPA goals for biodiversity, wildlife, habitat, and inferred landscape resilience. Thus, building upon existing guidance where connectivity objectives have been successful and determining a standardized science-based workflow to assess and prioritize connectivity that provides flexibility for local and regional mechanisms would greatly improve our ability to design functional landscapes.

1.1 Landscapes: the management canvas

Landscapes are the canvas in which ecosystem processes play out (Wiersma, 2022). Processes influence landscape patterns which in turn feedback on how processes behave (Turner, 1989). The field of landscape ecology was greatly spearheaded by how the interplay between disturbance and wildlife/vegetation responses result in emergent landscape patterns that drive ecosystem structure (Turner, 2010). The spatial patterning of habitat types and resources and their connectivity largely influence contagion processes i.e., animal movement (Kool et al., 2013), the flow and exchange of energy and matter (Shen et al., 2011), and the unfurling of different disturbance agents (Pearson et al., 2021). For instance, wildfire spreads across landscapes by consuming spatially connected fuel (i.e., dead vegetation, flashy graminoids, drought stress foliage…etc.), coupled with other drivers such as wind and topography (Bond & Keeley, 2005). Wildfire reduces fuel via consumption which in turn influences (1) the return interval of a subsequent fire event, (2) the regeneration of vegetation types based on post fire site conditions, and (3) rates of biomass accumulation back into fuel sources (Thompson et al., 2017). How the landscape is connected post disturbance influences where and how species and materials move between and among habitats/patches and in doing drives the spatial patterning of ecological communities i.e., source and sink processes (Heim et al., 2025). This example is also pertinent to other disturbances that alter contagion process and the distribution of ecological elements for example drought, overgrazing, and timber harvest. Understanding, the spatial patterning of landscapes with a focus on connectivity and how it underpins

ecosystem processes should be a fundamental consideration when developing management strategies at a landscape extent.

Management strategies for connectivity need to be measurable and verifiable. The quantification of landscape patterns can occur a-spatially via (1) composition and spatially via (2) configuration. Landscape composition is primarily an a-spatial assessment i.e., how much of a given feature is present. Landscape configuration is a spatially explicit assessment i.e., where a given feature is relative to other features in a matrix/mosaic (Cushman & McGarigal, 2019). Connectivity is inherently a spatially explicit assessment of landscape configuration (Tischendorf & Fahrig, 2001). In a forested landscape, patch and patch class scales are often used to assess connectivity e.g., patch informed by age and species composition or age and leading species (Spanowicz & Jaeger, 2019). These patch or stand elements are a common currency used among foresters, planners, resource specialists, and ecologists. Thus, the leap from patch/stand management to patch mosaic management that supports emergent connectivity patterns is achievable. However, concise management guidance paired with standardized analytical tools for consistent assessments is needed to inform effective management actions for landscape connectivity.

Increasingly, land managers are being tasked with developing management regimes that move beyond the stand to the landscape to deal with unprecedented disturbance events and more broadly the vast degradation of landscapes resulting in the loss of resilience. At these larger spatial extents, connected landscapes, are needed to support ecosystems services and resilience via processes while providing avenues of resource extraction to fuel economic systems at the forest stand scale (Harden et al., 2014). By considering this spatial hierarchy we can prioritize ecosystem properties at both scales to provide a better balance for ecosystem-based management. However, A central challenge here, is developing large spatial extent plans that encompass “big picture” motifs from ecosystem-based management frameworks including connectivity, corridors, and networks (Rudnick et al., 2012; Tarimo et al., 2025).

1.2 Connectivity: coarse and fine filter approach

In the same way that roads provide routes for the movement of goods, services, and people so do the pathways between nodes on a connectivity, corridor, or network map – the economic highways of ecology (Figure 1). The movements of fauna and flora through a landscape are often species specific with differences between hetero and conspecifics (Diniz et al., 2020; Koen et al., 2014). This provides additional challenges to design a functional landscape where connectivity is maintained for all species. Following the B.C. Biodiversity Guidebook (British Columbia, 1995) framework we can lump species

connectivity management using the coarse and fine filter approach (Figure 2). Species captured by the coarse filter are presumed to have afforded protection from OGMA's (Old Growth Management Areas) and Parks, along with old forests outside of OGMAs. For species that require fine filter management we would need to develop connectivity models based on habitat selection from GPS or other spatial positional data. By developing both coarse and fine filter management strategies for connectivity land managers can strive to strengthen, maintain, and restore the ecological connections that facilitate robust ecosystem economics.

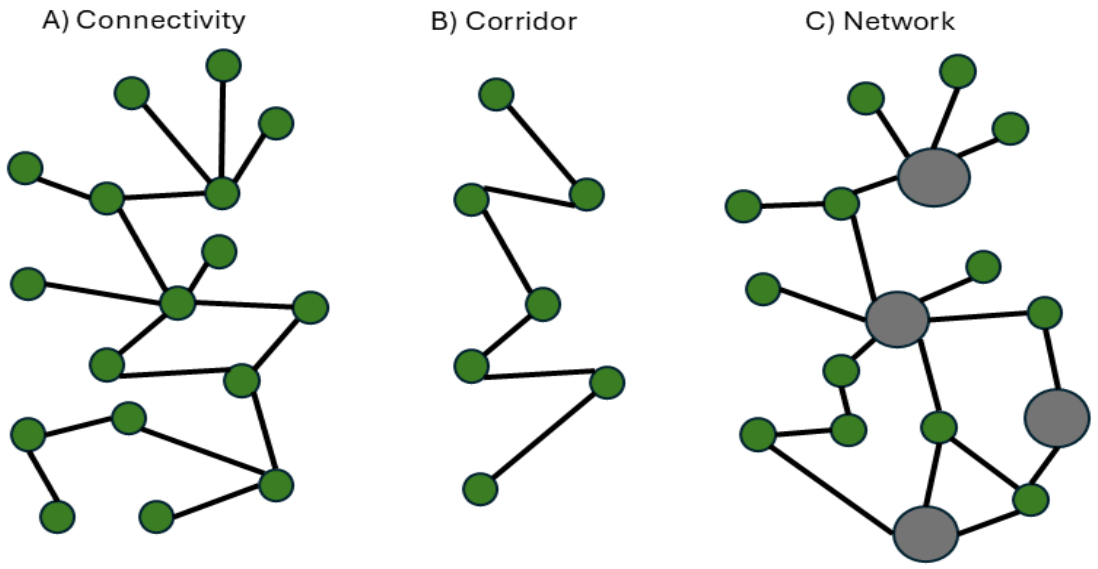

Figure 1. Concept diagram of A) connectivity, B) corridor, and C) network patch patterns. Connectivity (A) is shown as the connectedness of similar patches or patches that provide similar functionality and the pathways in which ecological information may flow through a landscape. A corridor (B) is largely a linear pathway that supports the directional movement of ecological information. A network (C) is a highly connected system with ecological anchors. These anchors are primarily conservation areas i.e., parks, protected areas, conservation lands – areas which are largely free from industrialized human disturbance. A network is connected outside of these anchors by features that support the movement of ecological information between, within, and among the anchors.

The coarse filter approach is low hanging fruit for implementing broad sweeping connectivity management in B.C. in the near term primarily within FLPs. Many multispecies connectivity models have identified the use of OGMA/old forests in movement between reserves and parks, with caveats that species specific models may be appropriate for some species (Brennan et al., 2025). Given the existing mechanisms for old growth management, priority deferral areas from the Old Growth Strategic Review (OGSR), and old forest seral assessments, many practicing foresters and land managers are familiar with handling these data. Thus, an additional analysis to conduct a connectivity assessment of old forests is a tangible request if tools and clear guidance are provided.

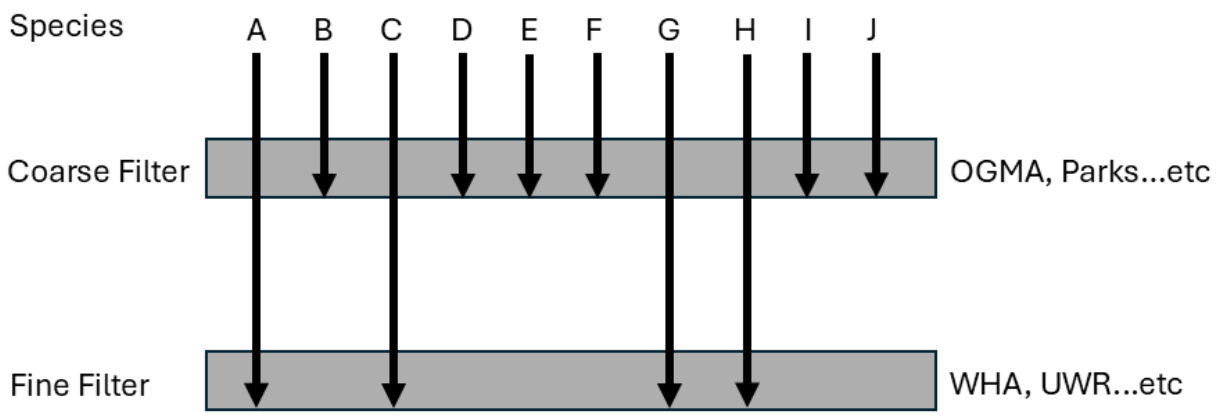

Figure 2. Coarse and fine filter approach to species management in BC following the BC Biodiversity Guidebook (British Columbia, 1995). The coarse filter captures species who are afforded protection from old growth management and Parks designation. The fine filter captures species who need specific protection or require nuanced habitat management. In the conceptual figure species B, D, E, F, H, and I are afforded protection by OGMA (Old Growth Management Areas) and Parks, whereas species A, C, G, and H require specific conservation management such as Wildlife Habitat Areas (WHA) and Ungulate Winter Ranges (UWRs).

1.3 Connectivity, biodiversity, and landscape resiliency

Resilience is the ability of a system to retain fundamental components that allow for recovery post disturbance (Allen et al., 2016). Recovery is integral to resilience (McWethy et al., 2019). Recovery rates, i.e., how fast a system rebounds post disturbance, is tightly linked to the movement and exchange of ecological information via ecological memory e.g., species presence and distribution in the local and regional environment for migration processes (Vellend, 2016) , assembly pathways driven by interactions that facilitate organization and coexistence, and site fidelity (Pearson et al., 2021; Peterson, 2002). Biodiversity hotspots are often beacons of ecological memory (Khalighi et al., 2022). These hotspots contain common to uncommon species and thus provide the ecological seeds necessary to encourage a system to return to a given state, i.e., resilience domain, through the legacy of species assemblages (Connor & Simberloff, 1979; Oliver et al., 2015). Landscapes composed of highly connected biodiversity reserves preserve ecological memory and have a higher potential to buffer state changes in response to disturbances and climate change (Keane et al., 2009). In general, connected ecosystems result in quicker recovery rates via ecological memory and the legacy effects of species interactions and movement patterns (Ogle et al., 2015). Thus, when designing functional landscape to build resilience, connectivity for biodiversity underpins crucial ecosystem processes that not only support recovery but resistance to disturbance.

Here, we present our findings from:

1. A comprehensive review of how land-use plans in British Columbia have defined, used, and implemented connectivity, corridor, and network management objectives.
2. A systematic literature review to identify potential connectivity modelling tools and their practicality for use within current Forest Landscape Planning initiatives
3. A modelling exercise using Omniscape to determine current connectivity from a coarse filter biodiversity lens i.e., forest intactness, for the Quesnel FLP.
4. A draft landscape management strategy for implementing landscape connectivity as an FLP objective.

Our overarching objectives for this project are to (1) identify the current management mechanisms and their implementation effectiveness for connectivity in existing and historical LUPs, (2) develop a science based review of connectivity tools and their practicality for implementation in a FLP, (3) implement connectivity modelling for the Quesnel FLP informed by the science review, and (4) construct a Values, Objectives, Indicators, and Targets (VOIT) framework for connectivity in the Quesnel FLP that leverages the LUP, science review, and modelled outcomes from the objectives1-3 listed above.

1.4 Data Availability

All data for this project can be found here:

https://github.com/Heckford/Cariboo-Connectivity.git

Additional information including Omniscape coding can be found here:

https://github.com/FOR-CAST/bc-connectivity.git

2. Phase 1: A comprehensive review of connectivity, corridor, and network management in B.C. land-use/strategic plans

2.1 Rationale for strategic/LUP review

Existing and historical landscape, resource management, and Land Use Plans (LUPs), hereafter collectively referred to LUPs, which set targets and objectives for connectivity, corridors, and networks may provide foundational information to build off. Currently there is no database or public repository to query and review existing and historical LUPs. This limitation restricts (1) temporal comparisons, (2) multi-plan objective assessments for effectiveness, and (3) data mining exercises to extract information. To address these limitations, with respect to connectivity focused management, we initiated a comprehensive review of existing and historical LUP's in B.C.

2.2 Methods: LUP review

To assess LUP's for connectivity, corridor, and network information we conducted internet searches and extracted all LUP's found within BC. For plans unavailable on the internet we sourced these documents by emailing companies, First Nations, or authors. For all LUP's we extracted information for connectivity, corridors, and networks. To ensure we captured corridors we also search for the term "connectivity corridors" for networks we searched for "networks", "forest ecosystem networks", and "functional ecosystem networks" since these terms are seemingly use interchangeably. In addition, we also extracted "wildlife movement" information if present. Data extraction followed the diagram shown in Figure 3. The LUP review database is available for public consumption here

https://github.com/Heckford/Cariboo-Connectivity/blob/85c9f335f802a02886322df72d9c7dcd8b92745b/Phase1_LUP_Review.csv

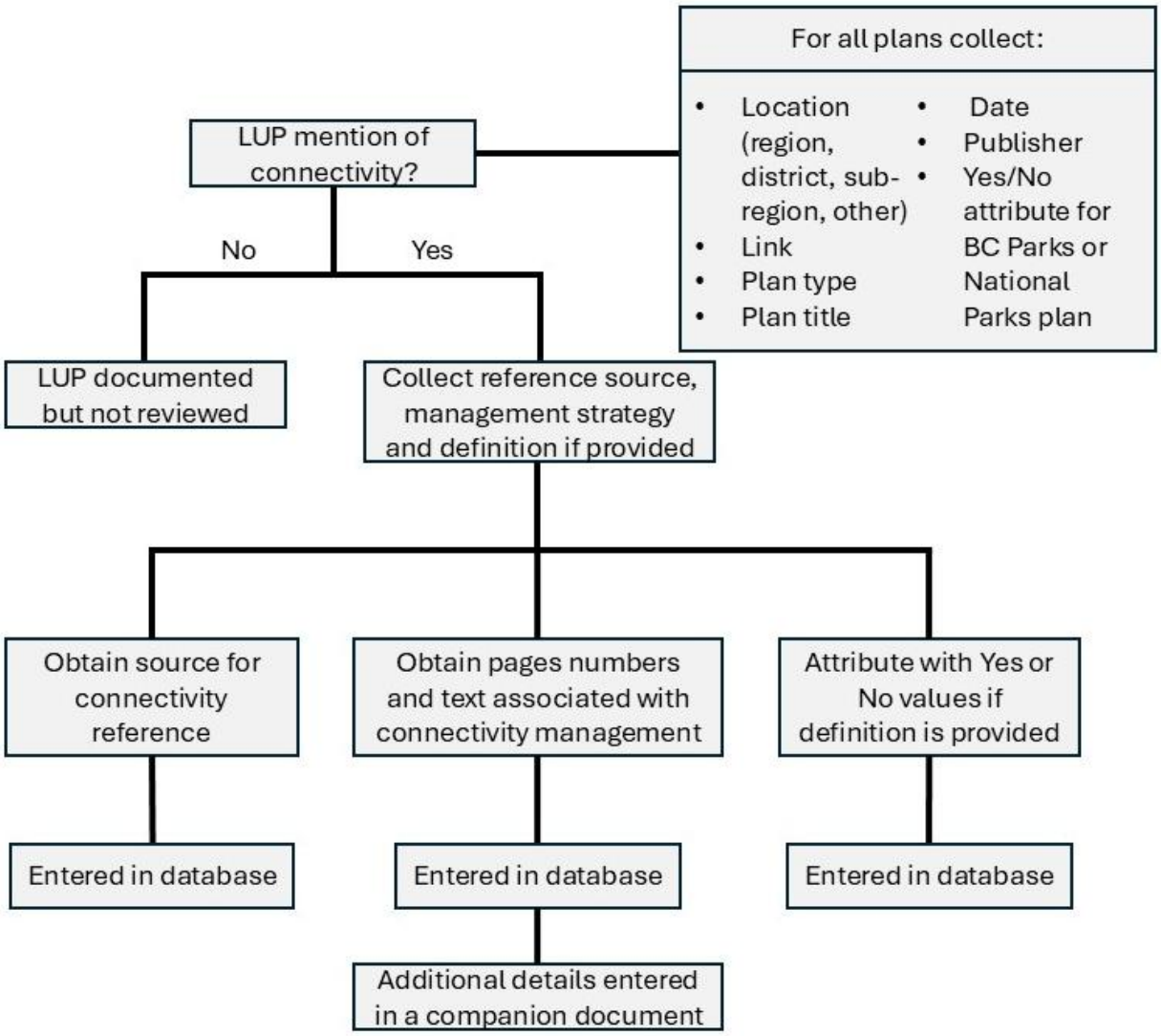

Figure 3. Flow diagram of how we extracted LUP connectivity information. For corridors and networks, we repeated the information extraction process. To ensure we captured corridors we also searched for "connectivity corridors" and to capture networks we also searched for "forest ecosystem networks" and "functional ecosystem networks."

2.3 Summary of findings

In total we reviewed 573 Land Use Plans and similar strategic plans including BC Park, heritage, and conservation areas, and National Parks (n = 4). Figure 4 shows the number of LUP's that (a) referenced, (b) linked to a management strategy, and (c) provided a definition for connectivity, corridors, and networks. We parsed these elements by B.C. Natural Resource Regions where applicable to develop a sense of the spatial distribution of connectivity objectives. In general, the south and west coast along with the Skeena region made larger contributions to connectivity objectives. However, out of the 573 LUP's we reviewed less than half referenced connectivity and corridors at 42% and only 13% referenced networks. Management strategies for connectivity, corridors, and networks were contained in 32%, 29%, and 9% of the reviewed LUP's respectively. Less LUP's provided definitions for connectivity, corridors, and networks at 5%, 3%, and 6% respectively.

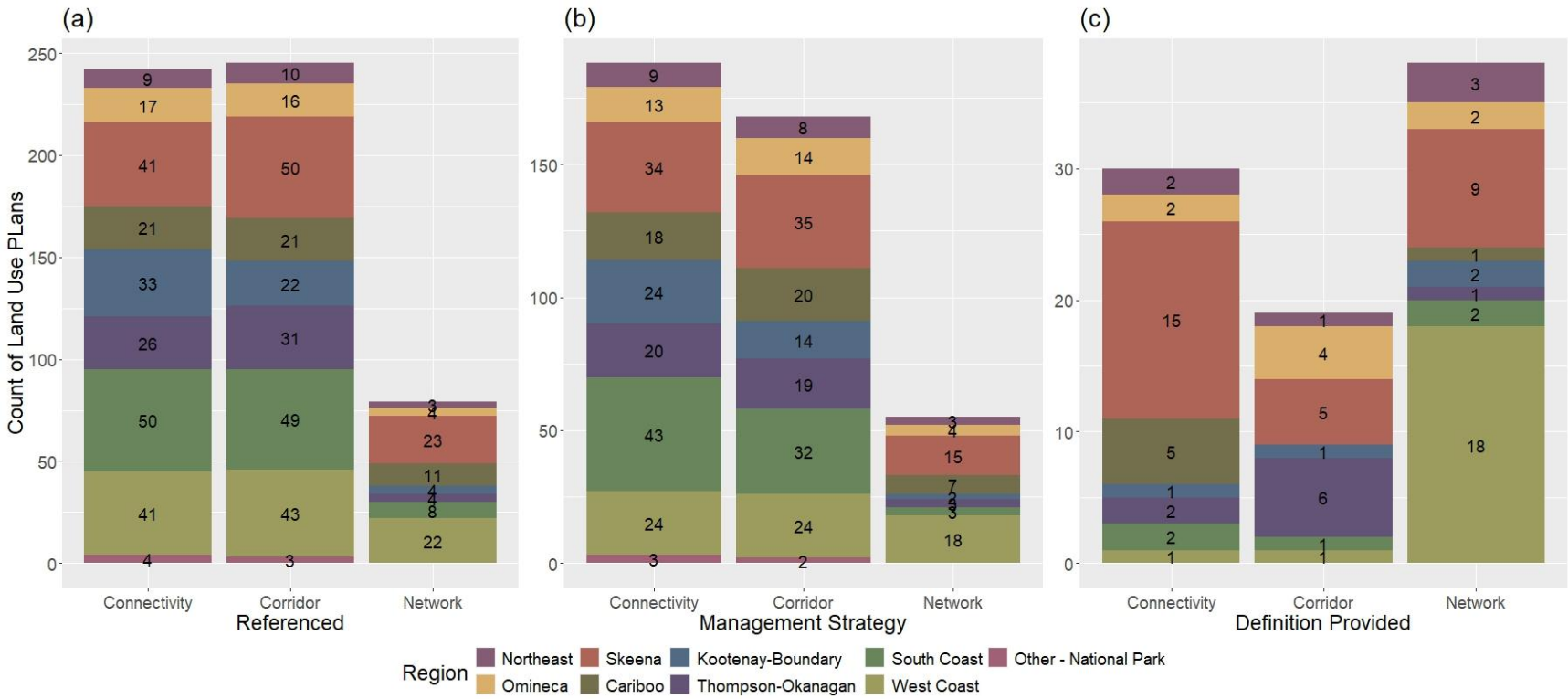

Figure 4. Counts of LUP's that (a) referenced, linked to a (b) management strategy and provided a (c) definition for connectivity, corridors and networks. We stacked the bar graph to show counts of LUPs by region – colour coded and ordered from north to south with other – National Parks noted as the last category.

To assess the vintage of the LUPs we reviewed we parsed regional plans by date (Figure 5). From Figure 5, we can see that the bulk of the LUPs we reviewed occurred around the year 2000's. This also coincides with an uptick in connectivity science which is discussed in Phase 2 (Figure 8 below).

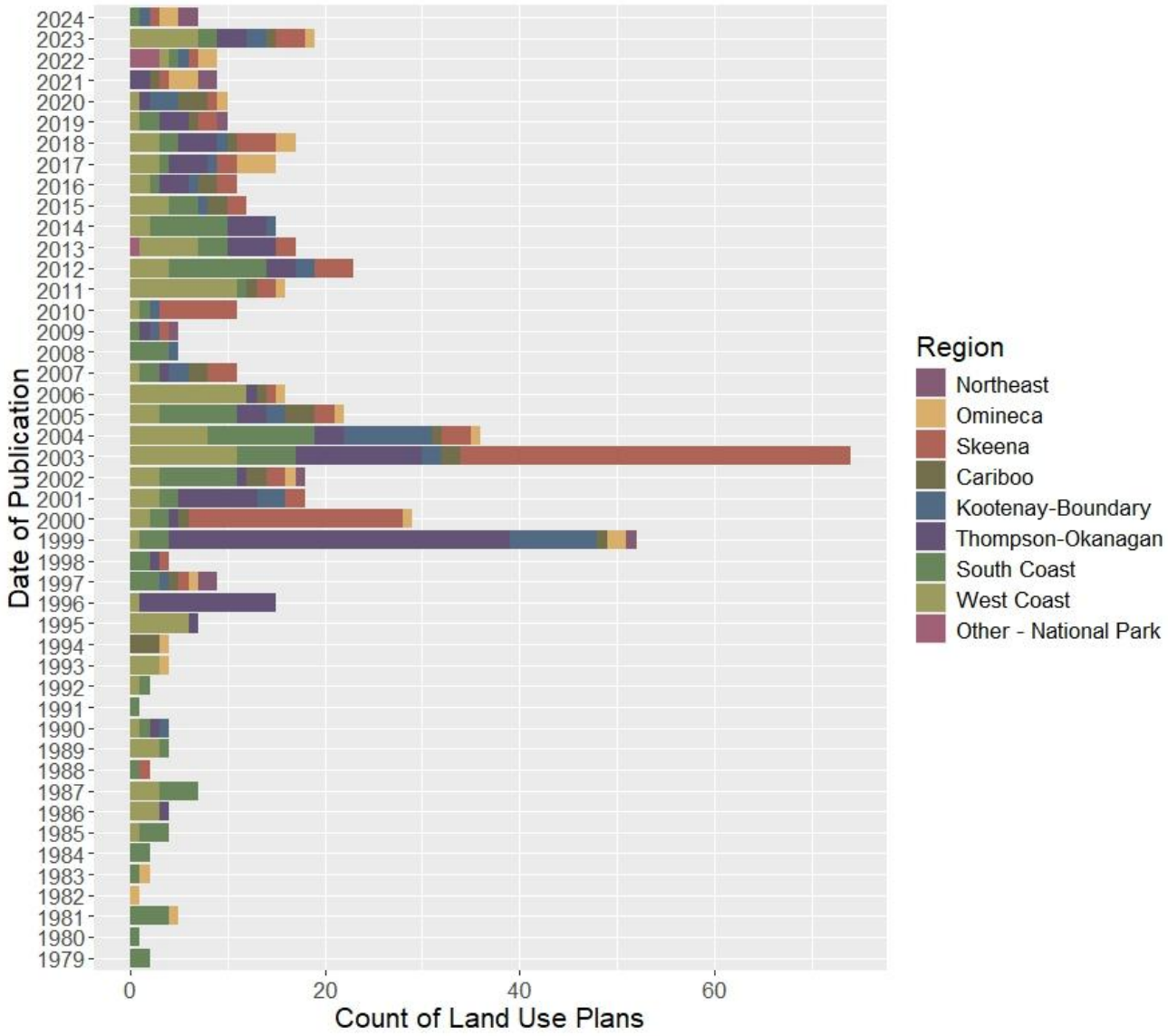

Figure 5. Assessment of LUP's review by date of publication shown by region.

In December 2024 we asked the Old Growth Working Group (OGWG) to review our LUP database and provide regional and local coments on (1) how practices for connectivity management were addressed, (2) what legal tools were used to implement practices and (3) additional information for connectivity, corridors, and network objectives (Figure 6). Of the 573 LUPs, 135 LUPs or 23% were reviewed by the OGWG for connectivity, 5% for corridors, and 2%

for networks. Of those only 53, 30, and 52 were confirmed to have practices implemented for connectivity, corridor, and network management respectively.

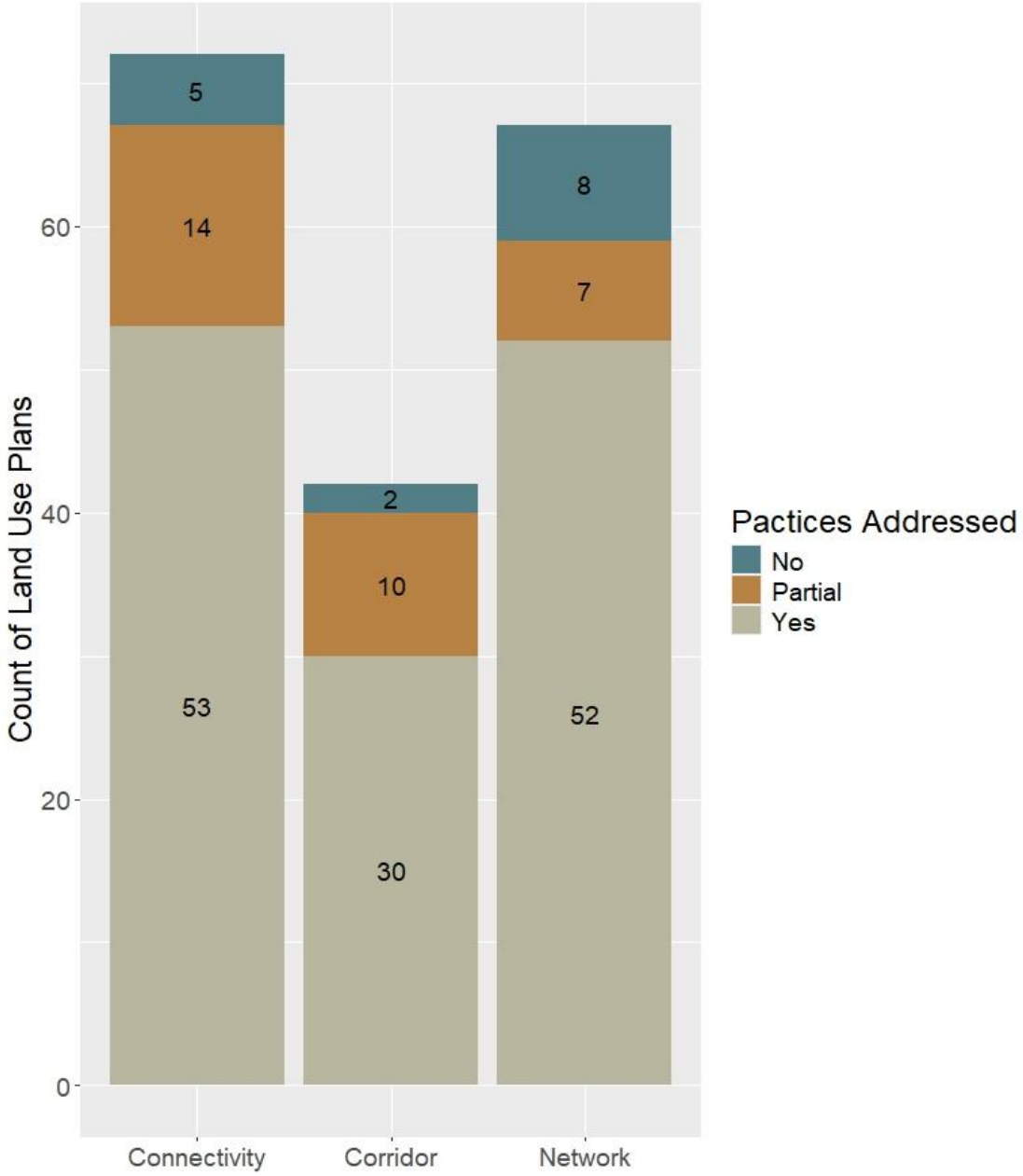

Figure 6. Counts of LUP's reviewed by the OGWG for connectivity, corridor, and network management to confirm practices discussed in the LUPs were addressed via management and objectives.

2.3.1 Interpretations and key findings

Our findings, informed by our review and OGWG comments, suggest management objectives for connectivity are poorly implemented in current LUPs within BC. This appears to stem from (1) a lack of understanding around the differences between connectivity, corridors, and networks, (2) vagueness in writing connectivity strategies that weaken developing verifiable and measurable objectives, and (3) non-descript quantitative pathways for assessing connectivity. For example, an objective which states "will maintain upland to lowland connectivity" does not articulate how this connectivity will be maintained and more importantly at which point is it not maintained i.e., which patch types provide connectivity, distances between those patch types, interior forest conditions…etc. There are many things to consider when delineating connectivity strategies and much of what was revealed in our review suggests a disconnect between the conceptual writing of connectivity objectives and the quantitative tools available to assess landscape connectedness.

2.4 Future work

We suggest future work may include (1) further review of LUP's not reviewed by the OGWG to strengthen or refute our interpretation of current connectivity management strategies, (2) review of LUPs in other jurisdictions i.e., other provinces, states, or countries, (3) finalize assessment of where connectivity, corridor, and network objective-based information is flowing from via a linkage assessment of foundational information. Addressing the third point would allow policy makers to focus efforts on where to provoke changes to certain foundational documents, i.e., the Biodiversity Guidebook, to encourage consistent and appropriate consideration of connectivity in land use and Forest Landscape Planning initiatives.

3. Phase 2: A systematic literature review of connectivity, corridor, and network publications

3.1 Rationale for literature review

A key gap highlighted by our review of LUPs in Phase 1 was the disconnect between writing connectivity strategies that link to tools and quantitative pathways for assessing connectivity. In Phase 2, we conducted a systematic review of the literature to (1) understand the historical development of connectivity science, (2) derive contemporary tools, and (3) evaluate the practicality of these tools for land managers.

3.2 Methods: Literature review

We used Web of Science to extract peer reviewed literature for connectivity, corridor, and ecosystem network science. We focused on literature which contained two key elements (1) a tool or software and (2) management outcomes. We used the following strings to extract literature:

- TS=(Connectivity) AND (Ecolog*) And (Tool OR Software) AND (Management)
- TS=(Corridor) AND (Ecolog*) And (Tool OR Software) AND (Management)
- TS=(Network) AND (Ecolog*) And (Tool OR Software) AND (Management)

Initially we set the start date to 1960 but later constrained the search window to 2005-2024 to capture the last 20 years of scientific work for PDF screening. We extracted all returned literature as PDFs and compiled PDFs into a Zotero open-source repository. We followed PRISMA standards which were designed to increase the transparency of systematic literature reviews via reporting why the review was done, what the authors did, and what they found (Page et al., 2021). We provide the PRISMA diagram below describing our approach (Figure 7).

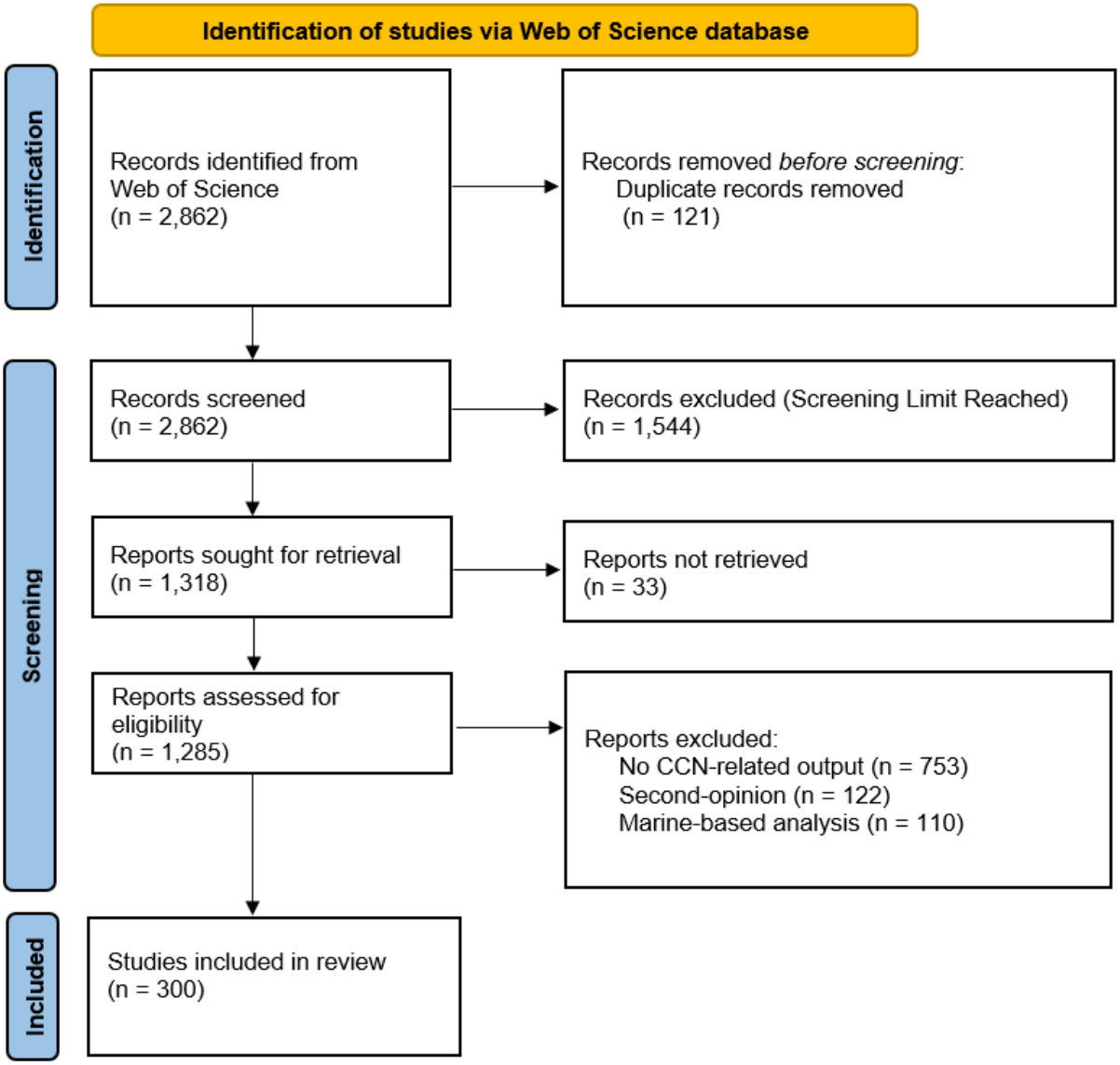

Figure 7. PRISMA diagram outlining our systematic literature review. The use of CCN here denotes Connectivity, Corridors, and Networks. The diagram summarizes the total processing of literature to produce a sample size of 300, i.e., 100 papers used per each connectivity component – connectivity, corridors, networks.

Our three queries returned a total of 2862 papers. We subsequently removed duplicates, screen records and subtracted reports that we were unable to retrieve, resulting in final tally of 1285 publications for eligibility assessments. From those publications we further screened if they (1) had no connectivity, corridor, or network related output, (2) focused on marine systems, and (3) expert second opinion on the applicability of the publication to contribute to our assessment for LUP connectivity management. The primary objective of our screening was to produce a sample size of 100 publications for review of management tools for each connectivity element. To do so, we pulled the 100 most recent papers

starting in 2024 and working back in time for each query i.e., 100 publications for connectivity, 100 publications for corridors, and 100 publications for networks. When screening these 100 publications we rated their suitability as either Yes, No, or Maybe. For papers rated No, we pulled the next publication from the candidate pool i.e., publications that were successfully assessed for eligibility. This ensured that if we exclude publications, other publications next in line by date would replace it to allow us to achieve a 100-publication sample size. From these 100 publications we collected the following information in a database:

(1) Bibliographic details: year, title, author, DOI, and abstract
(2) Analytical tools used
(3) Models
(4) indicators
(5) Data used and sources if provided
(6) Scale of the analysis, including study scale, scale considerations, multi-scale approaches, and study area
(7) Whether the study was terrestrial- or aquatic-based; if aquatic, freshwater or marine was recorded
(8) Whether the study was species-agonistic- or species-specific-based.
    8.1. If species-specific we record the species group studied e.g., mammals, birds, reptiles
(9) Whether the analysis was CCN-specific or conceptually similar

The literature review database is available for public consumption here https://github.com/Heckford/Cariboo-Connectivity/blob/85c9f335f802a02886322df72d9c7dcd8b92745b/Phase2_Literature_Review.xlsx

From the literature review we extracted lists of (1) analytical tools, (2) models, and (3) indicators. We transposed these data into list format and evaluated the uniqueness and frequency at which these elements occurred in our publication sample. From these outputs we extracted the top 5 tools. We reviewed these tools for their application to assist land managers with connectivity modelling to support setting objectives that are measurable and verifiable. Of the top 5 we identified the tool which is primarily used by researchers that balances computational complexity, practical use, appropriate scales of application, and translatability to objective setting.

3.3 Summary of findings

In total our three Web of Science queries returned 2862 publications. Although we constrained our screening to the last 20 years, the trends of connectivity science from 1980 onwards suggest a major uptick in scientific development in the early 2000's (Figure 8; Liczner et al., 2024). The emergence of computer processing systems that can handle large spatial data in the early 2000's likely contributed to the advancement of connectivity science. The curves in Figure 8 are not cumulative, the counts are publications by year, and the trends suggest that network science by far outweighs the annual contribution to connectivity science. That said, the publications captured here span across scientific disciplines – however ecological connectivity merges and uses concepts from a variety of disciplines (Liczner et al., 2024).

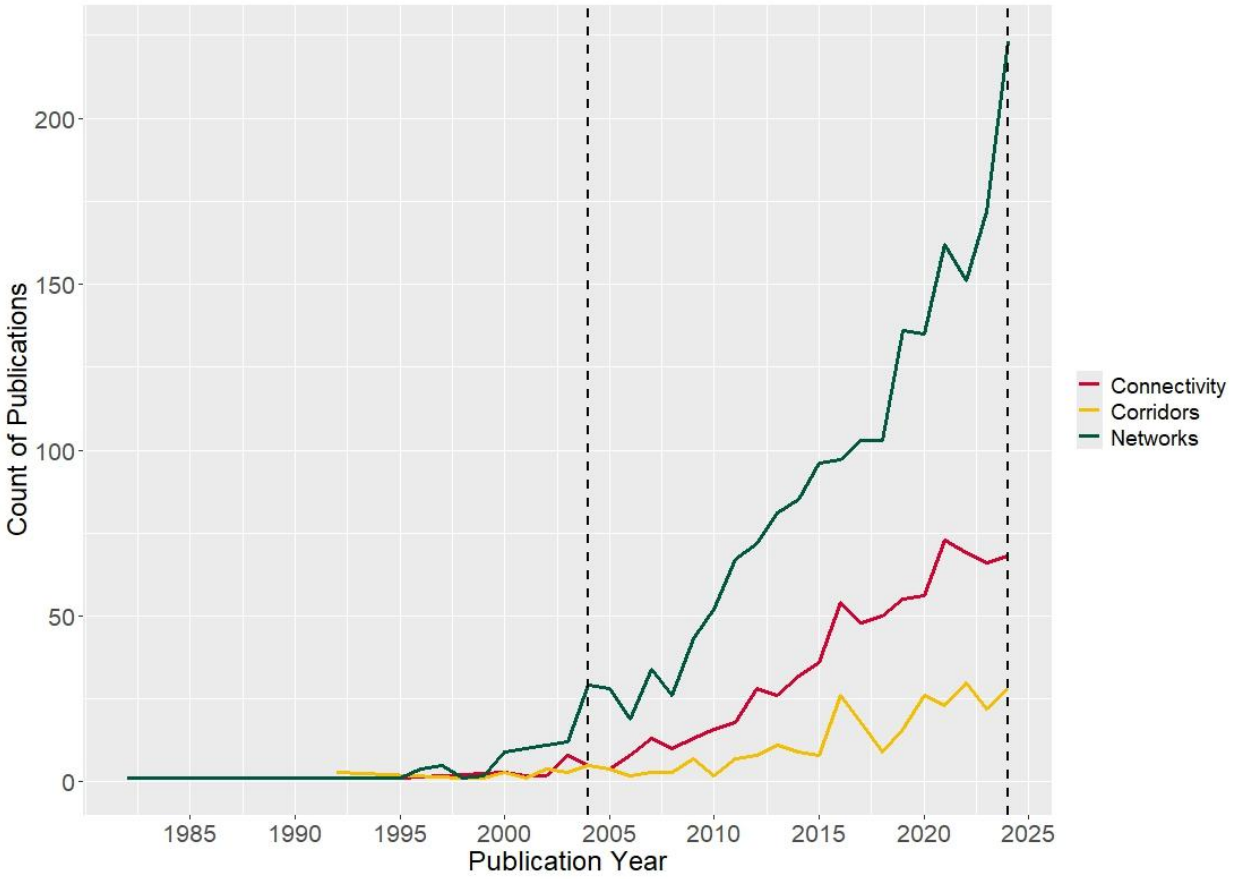

Figure 8. Count of literature for connectivity (red line), corridor (yellow line), and network (green line) publications from 1980 to 2024. The vertical dashed lines show the timeframe from which we screen publications for use in our review of analytical tools for connectivity land use planning.

The central focus of our literature review was to identify potential tools that provide an accessible platform at the practitioner skill level for implementing connectivity

assessments via modelling that translates to LUP objectives. Our review provides a snapshot into plethora of current analytical tools (Figure 9). However, upon review and in discussion with subject matter experts it became clear that although certain tools are widely used for scientific inquiry, the practicality of those tools for land use planners and the scale at which those tools can provide information narrowed the choices down to a few.

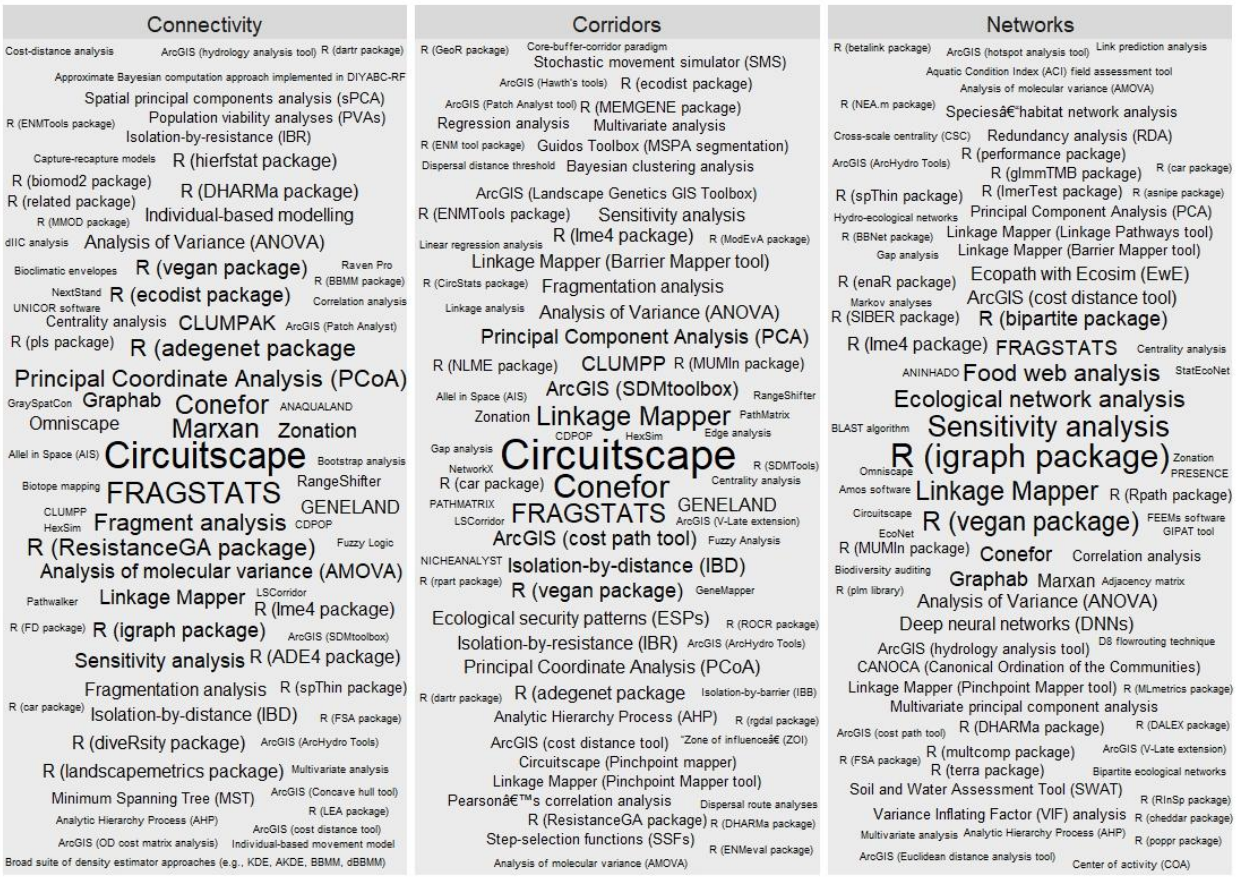

Figure 9. Word cloud of tools with size by frequency of occurrence derived from our screening of 100 publications for connectivity, corridors, and network science.

3.4 Recommendations for connectivity modelling

Although many tools are available for connectivity modelling (see Figure 9) very few offer the accessibility needed to encourage connectivity modelling and objective setting in land use planning specifically Forest Landscape Planning initiatives. Omniscape, an implementation of Circuitscape to model omnidirectional patterns, emerged as a clear candidate given (1) accessibility via documentation, (2) open source code, (3) numerous peer reviewed examples, (4) scalability, (5) replication potential, (6) ability to perform species-specific and species-agnostic assessments, and (7) demonstrated uses in various systems (Anantharaman et al., 2020; McRae et al., 2008, 2016).

3.5 Future work

Future steps could be taken to strengthen our interpretations of the literature review findings including (1) fine tuning the assessments beyond the 100-publication sample size, (2) further cleaning of the tool, model, and indicator frequency databases to determine clear trends in tool emergence and usage in scientific pursuits, and (3) use those frequency databases to extract those top ranked tools and develop compare and contrast assessments for a comprehensive understanding of tool commonalities and differences.

4. Phase 3: Implementation of Omniscape to model connectivity for the Quesnel FLP

4.1 Introduction

Landscape connectivity is instrumental for ecosystem recovery (Damschen & Brudvig, 2012; Standish et al., 2014), shaping the distributional patterns of ecological elements i.e., species distribution and population densities (Belote et al., 2022; Koen et al., 2014), and driving feedback processes such as fire spread, nutrient flows, and genetic movement (Hutchinson, 1948; Pichon et al., 2024). As a staple for wildlife and biodiversity management (Bacar, 2025), and as a tool to ameliorate climate change effects and ecosystem transition, landscape connectivity is a crucial factor to consider in landscape planning (Bedson et al., 2025). Our work in Phase 1 demonstrated (1) a lack of consistency in the use of connectivity and connectivity elements i.e., corridors and networks in current land use planning, (2) a clear failure to develop robust connectivity objectives that are legally binding, (3) inadequate consistency via analyses to support measurable and verifiable objectives, and (4) hampered decision-making that limits an understanding of impacts furthering the effects of landscape degradation (Karlsson & Bodin, 2022). Developing landscape connectivity objectives is a key component to managing landscape resilience (see infographic in Appendix 1) and as such should be a central focus from which other landscape planning components stem i.e., timber, recreation...etc. As described above, connectivity objectives in LUP/FLP and other strategic plans could either be species-specific or species-agnostic. Species-agnostic approaches should underpin biodiversity principles and align with existing biodiversity management strategies and tactics. In Phase 2 we identified Omniscape as an accessible and practical tool for entry level practitioners to implement, develop, monitor, and assess landscape connectivity. Here we provide an example of how to implement species-agnostic connectivity modelling using Omniscape for the Quesnel Natural Resource District (DQU).

4.2 Methods

4.2.1 Study area

Our study area is the Quesnel Natural Resource District located in central British Columbia. The DQU is in the process of developing a FLP and as such an ideal case study to support connectivity objective setting via modelling. The DQU is approximately 20,833km^2 extending from the Ilgachuz Range in the west to the Cariboo Mountains in the east covering eight different biogeoclimatic zones (Figure 10). We buffered the study area by 500 pixels at 30 m to reduce the erroneous edge effects when modelling connectivity patterns.

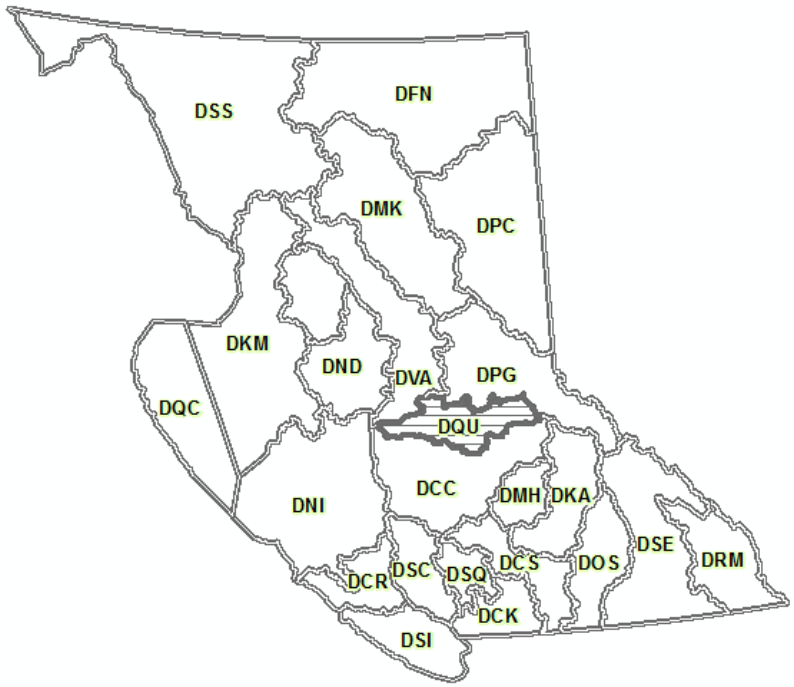

Figure 10. Map of the Quesnel Natural Resource District (DQU) relative to other natural resource districts.

4.2.2 Omniscape overview

Omniscape uses a moving window analysis that applies Circuitscape iteratively in a user defined radius. It requires three inputs, (1) a moving window size, (2) a resistance raster and (3) a source weight raster. The resistance raster defines the traversal cost of pixels in the moving window analysis. The source weight raster defines the relative amount of current that a given pixel equates to. We use 'current' here to describe the resistance at which electrical flow is impeded. Pixels with a source strength greater than 0 in the window are referred to source pixels. Within the window the target pixel is set to ground, and the source pixels are set as current sources. The total amount of current equates to the source strength of the target pixel and is divvied between the source pixels proportional to a given

pixels source strength to determine the current flow between source and target pixels (McRae et al., 2016).

4.2.3 Nearest neighbour analysis

A critical step for running Omniscape is determining the moving window size. There is no one size that fits all answer. The moving window size, i.e., radius, should be set to an ecological meaningful radius depending on (1) the hypothesis or (2) the objective of the modelling. Our approach here is to model species-agnostic connectivity via old forest connectivity. To determine our moving window size, we used a nearest neighbourhood analyses to compute the patch edge to edge distances between old forest patches (Ohmann et al., 2012; Pommerening et al., 2020). Our approach is ecologically meaningful in that we use the spatial distribution of old forest patches to inform a search radius for detecting connectivity between old forest patches. We determined seral stage using definitions from the Biodiversity Guidebook and patch using definitions from the Forest Biodiversity Cumulative Effects Framework (2020). We joined Simple Inferred Forest Age (SIFA) from the Forest Disturbance layer with the Vegetation Resource Inventory (VRI) and partitioned age classes using the Natural Disturbance Types and Biogeoclimatic (BEC) zones old classifications to determine patches. Using existing python code from the Cariboo Region Cumulative Effects Specialist we implemented patch definitions in R (R Core Team, 2020). Figure 11 below shows a simplified process of determining patch and Figure 12 shows the resulting seral patch distributional patterns for the DQU. From our nearest neighbourhood analysis, we selected the 20th percentile value i.e., 45.2 km, as the maximum search radius and the 100th percentile value of 1.7 km as the minimum search radius (Figure 13).

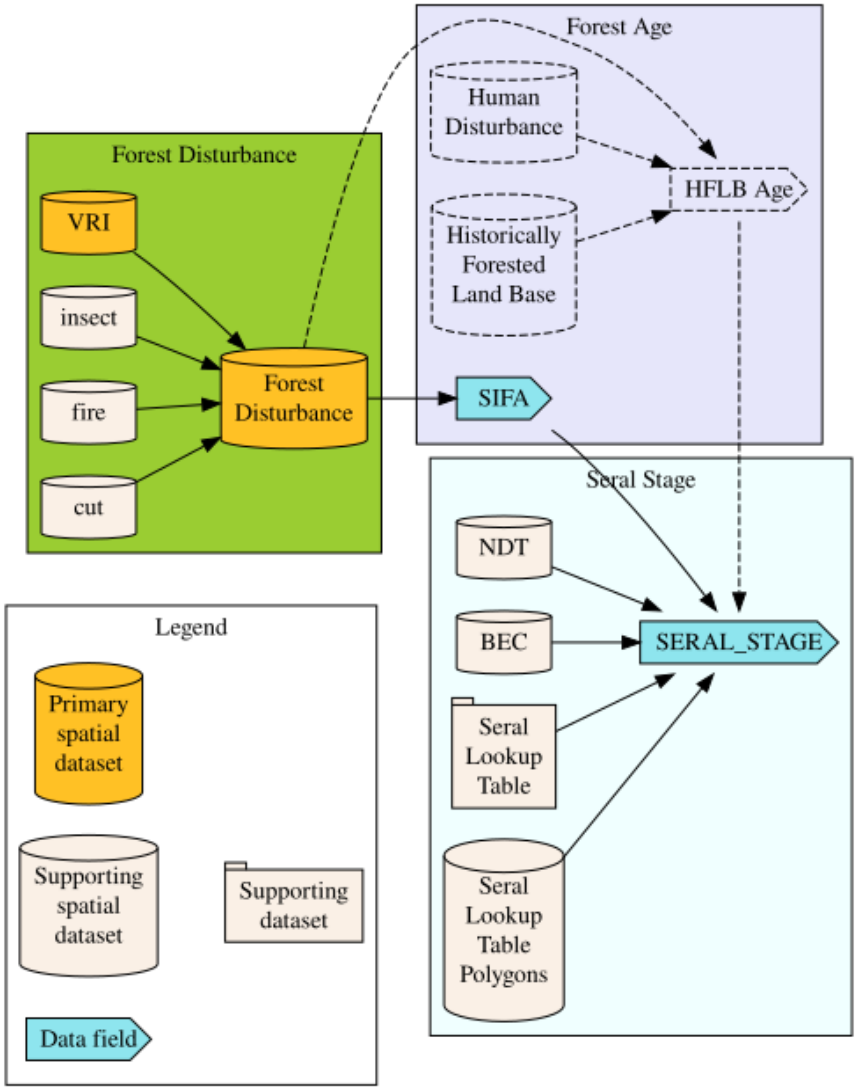

Figure 11. Determining patch type and size by BEC and NDT for leading species and age following the Biodiversity Guidebook, the Forest Biodiversity Cumulative Effects Framework and legal requirements from the Cariboo-Chilcotin Land-Use Plan.

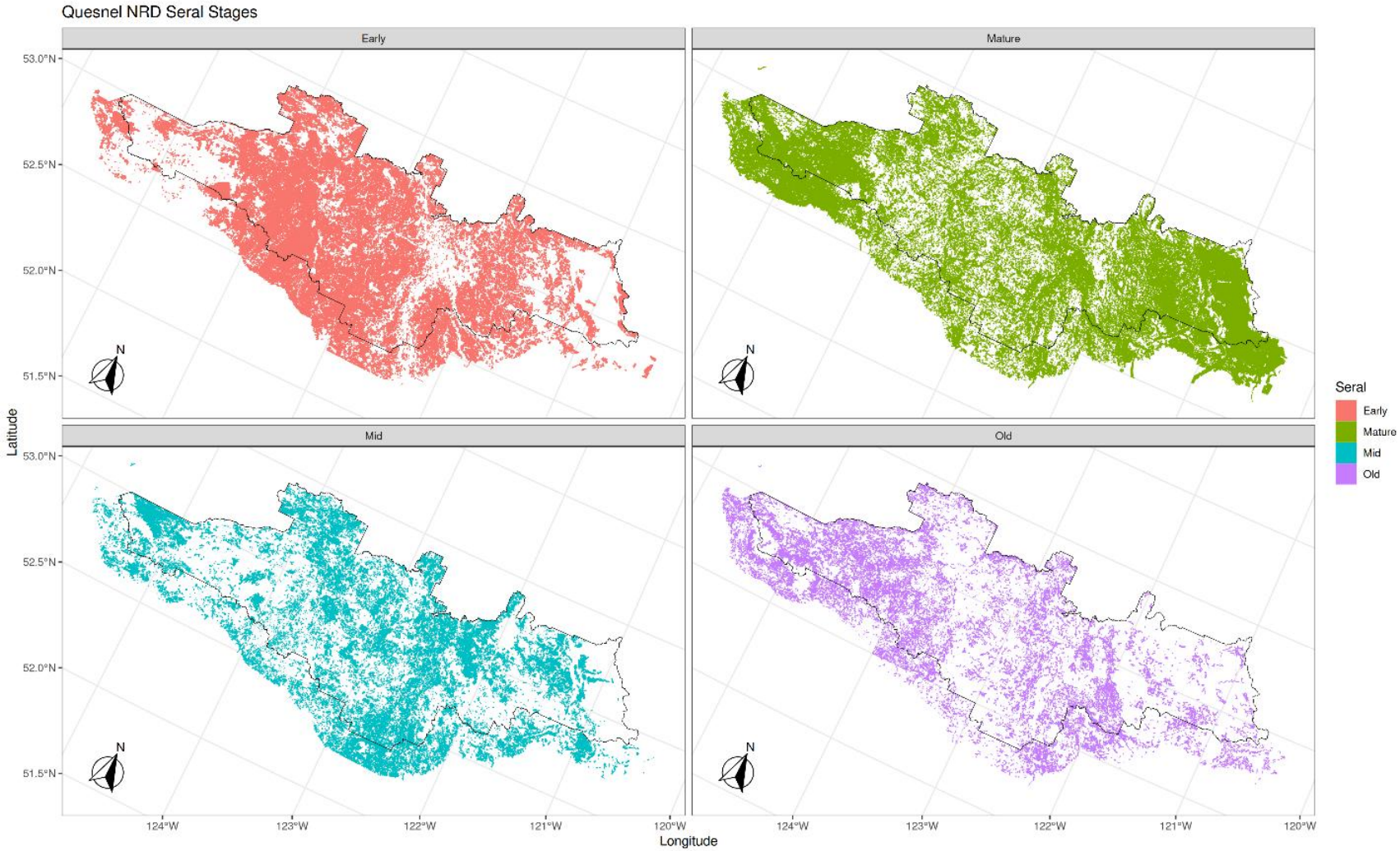

Figure 12. Seral and patch distribution patterns for the DQU.

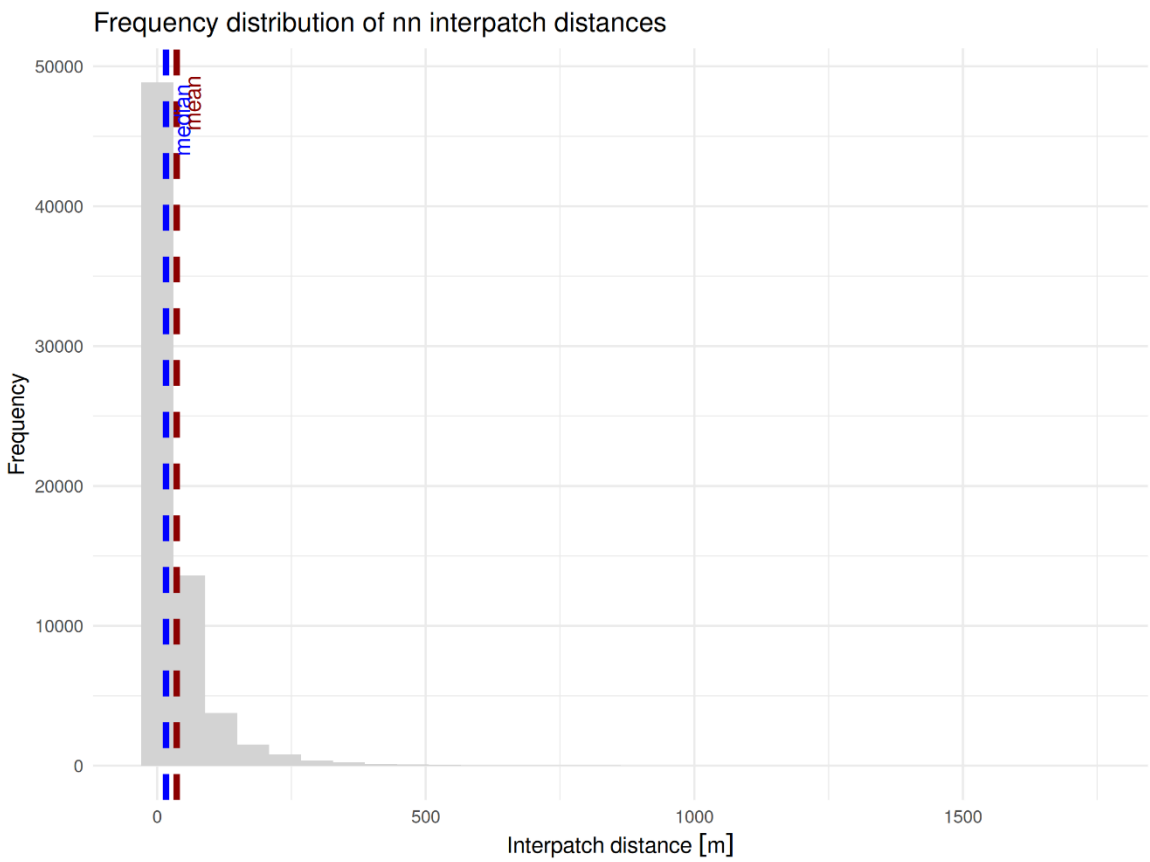

Figure 13. Distribution of nearest neighbourhood (nn) distances. The distribution is left skewed. We decided to use the 20th percentile as it captures the medium and mean or central tendency of patch distributions and the 100th percentile to ensure we captured all potential distances.

4.2.4 Data preparation

We assigned the range of resistance values from 1 to 1000 where a low value equates to high biodiversity i.e., older forests; a high value represents low biodiversity i.e., young forests. Conversely, we assigned the range of source weight values from 0 to 1 where a low value equates to poor habitat potential. We rasterized all data at a 30 m spatial resolution to align with the national 2020 landcover raster. We prepared all data using the terra package (Hijmans et al., 2020). We developed a spreadsheet to document (1) data used, (2) their corresponding resistance and source weight values and (3) rationale for those values from peer review landscape connectivity literature, we also captured (4) data buffering and (5) filtering if applicable. Our data treatment documentation is available here:

https://github.com/Heckford/Cariboo-Connectivity/blob/85c9f335f802a02886322df72d9c7dcd8b92745b/Phase3_Omniscape_DataWeighting.xlsx

In general, given the four seral stage classes we partition resistance and source weight by quarters. For old forests we assigned a resistance and source weight value of 1. In comparison, we assigned a resistance value of 750 and a source weight value of 0.25 for young forests. We assigned wetlands a moderate resistance of 250 and source weight value of 0.75. We assigned lakes/water bodies to a maximum resistance of 1000 and source weight of 0. For streams we assigned small streams lower resistance and larger streams higher resistance all streams were assigned 0 for source weights. For linear transportation features we assigned varying resistance and source weight values depending on the type of transportation attribute and its impact to biodiversity connectivity i.e., highways have a larger impact than resource roads.

We combined all resistance rasters to create a composite resistance raster using the following rules:

1) Roads and water features can only increase resistance, not decrease it;
2) Final composite raster was reclassified to avoid 0 values (set to 1) and fill NA values as high resistance i.e., 1000.

We created a composite source weight raster using inverse logic, following these stacking rules:

1) Roads and water features can only reduce source weights, never increase it;
2) NA values were replaced with 0 to ensure compatibility with Omniscape.

4.2.5 Implementing Omniscape

We used Omniscape to produce connectivity maps. Our data preparation workflow generated several Omniscape configurations by adjusting the following:

1) Pixel size (p): either 30 m or 90 m; smaller pixel size increased computation time but did not produce qualitatively different results than larger pixel sizes;
2) Radius (r): the moving window size (in pixels), based on the results from the nearest neighbourhood analysis; the smaller r values correspond to the 20th percentile and the larger r values corresponds to the 100th percentile related to the pixel size used;
3) Block size (bs): used to speed up computations, set approximately to 10% of r (in pixels).

These conditions resulted in the following outputs:

Table 1. Potential Omniscape outputs by pixel size, radius, and block size.

Omniscape configuration	Description
2026-01-13_p30_r1802_bs181	30m pixels; radius 1802 pixels; block size 181 pixels
2026-01-13_p30_r183_bs19	30m pixels; radius 183 pixels; block size 19 pixels
2026-01-13_p90_r601_bs61	90m pixels; radius 601 pixels; block size 61 pixels
2026-01-13_p90_r61_bs7	90m pixels; radius 61 pixels; block size 7 pixels

For practitioners we recommend using a pixel size of 90 m and a radius size of 61 pixels (smaller value) as it provides a decent resolution in data while balancing computational time for an output that is evidence based and easily interpretable for planning purposes.

4.2.6 Post processing map making

Omniscape produces resistance-based flow outputs. We characterize these scores into the four following categories following justification from Schloss et al., 2022. The interpretation of these values is provided in Table 1:

Table 2. Description of resistance scores for connectivity interpretation.

Score	Name	Description	Implication	Management action
≤0.2	Limited Movement Potential	Land types (e.g., cliffs, open water), uses (e.g., intensive harvesting) or infrastructure (e.g., towns, major roads) restricts movement	Movement is low due to high resistance which could be natural or anthropogenic	These areas do not contribute to connectivity.
0.2 - ≤0.7	Impeded Flow	Decreased flow – movement options are limited	Movement is restricted due to fragmentation from roads, development or intensive land use. Movement is possible, but inefficient.	These areas are high priority for restoration actions (i.e., road deactivation, habitat enhancement). Limit additional devel opment.

>0.7 - ≤1.3	Diffuse Flow	Neutral flow with many connectivity options and are the most common.	Landscape is permeable with few major barriers or bottlenecks.	Maintain heterogeneity, appropriate patch size and seral stage distribution to preserve permeability.
>1.3 - ≤1.7	Intensified Flow	More flow than expected, reduced connectivity options;	Broader corridors than channelized flow which often provide movement routes through heterogenous terrain.	High priority for conservation or minimal disturbance.
>1.7	Channelized Flow	More flow than expected, where connectivity options are limited to as little as a single route.	Critical connectivity pinch points. Losing or degrading these would disproportionately damage connectivity.	High priority for conservation or minimal disturbance

4.2.6 Code availability

Our code and additional results are available on GitHub here: https://github.com/FOR-CAST/bc-connectivity?tab=readme-ov-file#overview

4.3 Results

Below we present results for the 90 m pixel size and 61-pixel radius moving window size. We characterize resistance scores into the five categories described in Table 2. In Figure 14 we show the connectivity results categorized in the 5 classes for the entire DQU. In Figure 15 we show a finer scale example for two landscape units with connectivity features overlapped with land designations to demonstrate how land managers may leverage existing or potential protection/conservation designation to support, retain, or restore connectivity in degraded landscapes.

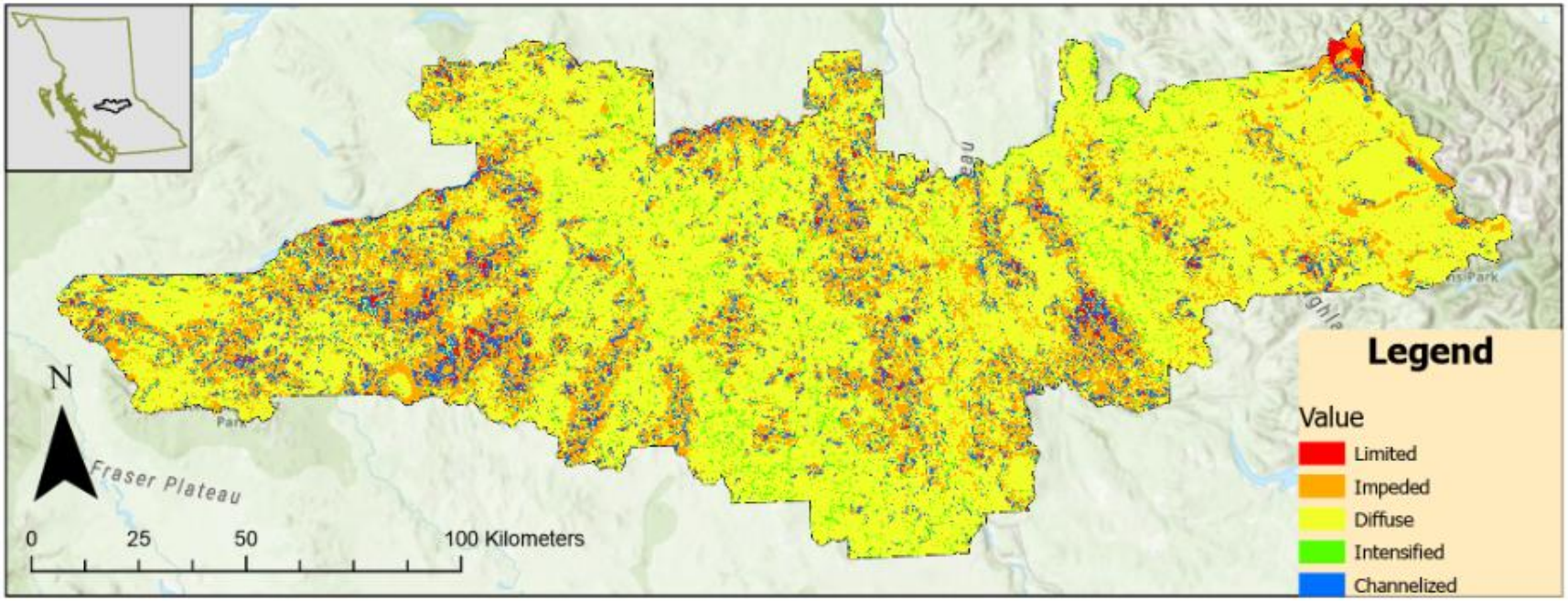

Figure 14. Omniscape output for the DQU showing connectivity characterized by resistance score classes (Schloss et al., 2022).

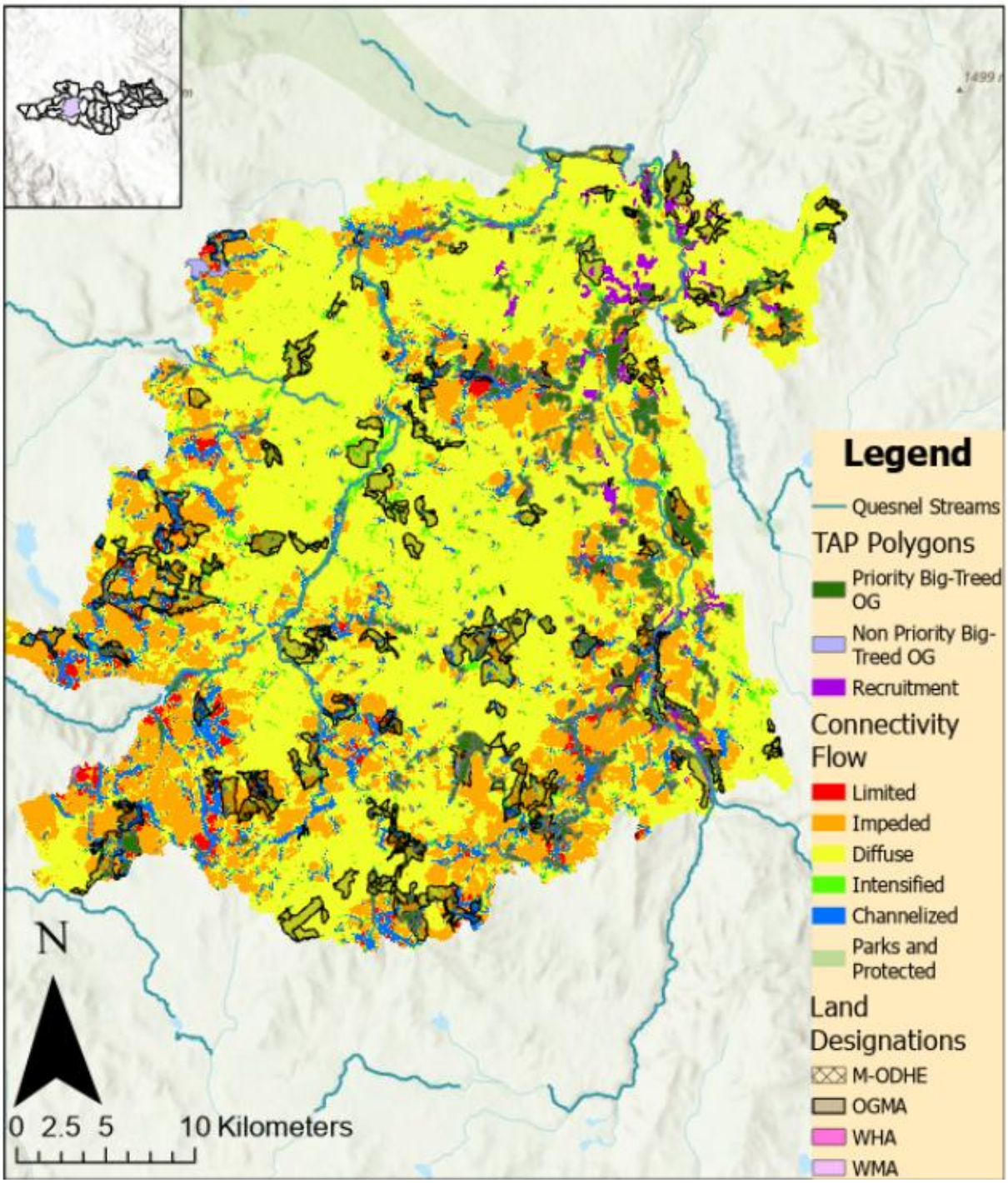

Figure 15. Baezaeko and Marmot Landscape Units within the DQU showing connectivity output overlayed with land designations and old growth deferrals as a management example for planning purposes.

4.4Discussion

Omniscape offers a versatile and practical approach for entry level individuals to model landscape connectivity. Given the information needed to implement the modelling environment, planners can easily conduct modelling and align those modelling conditions with written objectives that provide measurable and verifiable statements. Moreover, the code we have developed here can be transferred to other regions or adapted to input locally and regionally specific data. The characterization of connectivity outputs in the 5 categories (see Table 2 above) provides a translatable description of landscape objectives and provides foundational information to construct recovery, maintenance, and enhancement management strategies. For example, in areas of diffuse or impeded flows, managers could aim to strengthen connections between intensified and channelized areas by implementing objectives that recruit, maintain, or enhance old growth conditions. Furthermore, managers could set independent objectives for connectivity, corridors, and networks (see differences in Figure 1 above). The maps produced by Omniscape allow us to focus on areas to strengthen these connectivity components. For example, along rivers, streams, and wetland complexes we may have discrete patches of channelized, and intensified flow, peppered with diffuse and or impeded flow. Here we can alter biodiversity elements to support connecting channelized and intensified flow to result in corridor potential. We can also use these areas as ecological anchors to support bridges to other soft protected areas that encourage connectivity and network development.

4.5 Conclusion

Omniscape is a powerful yet accessible tool that allows practitioners to model both species-specific (modelling approach will need species data) and species-agnostic connectivity – as demonstrated in this example. Species specific modelling will require additional steps to inform data priors of connectivity via Resource Selection Functions (RSF; Boyce et al., 2002; Chetkiewicz & Boyce, 2009). We recommend practitioners use a resolution (i.e., 90 m) that captures the scale of input data while balancing for computational limitations. We found that a 90 m pixel size produces suitable results that were comparable to our 30 m pixel size outputs but at a much lower time and computational processing cost. The outputs from Omniscape are easily interpreted when characterized in the 5 flow categorizes described above (Table 2). These characterizations allow planners to implement strategies to recover, maintain, or enhance connectivity that easily translate to empirically informed measurable and verifiable objectives in an FLP or LUP document.

5. Phase 4: Connectivity VOIT for the Quesnel FLP

The FLP team in the DQU is using the VOIT (Values, Objectives, Indicators, Targets) approach to developing landscape management strategies. Below we have attempted to compile our learnings from Phase 1, 2 and 3 to inform a potential VOIT.

Table 3. Example VOIT for the DQU FLP to inform landscape connectivity tied to biodiversity i.e., species-agnostic approach. The table below is not complete but provides a launch pad for FLP discussions to support species-agnostic and biodiversity focused connectivity management.

What is important to manage, conserve or protect?	
Value	Landscape connectivity
Sub-Value(s)	Old forest management
What do we want to achieve and what are the current requirements?	
Objective	Maintain, enhance, and recruit connectedness among old forest patches, protected areas and other conservation focused management zones to support connectivity, corridors and ecosystem networks.
Existing legal requirements	**None**
Current Management Strategy	**Some precedent with FSPs but assessment and consistency are lacking**
What is the desired result?	
Outcome	1. Implement connectivity management regimes: • Scale could be BEC, Landscape Unit, or other if the rationale is ecologically meaningful. 2. Compare connectivity model outputs against HRV and FRV model outputs: • Depending on ecosystem trajectories algin connectivity management plan to reflect HRV and or FRV. • HRV and FRV provide benchmark estimates of the spatial arrangement for old forest patches and connectivity between and among them.

	3. Using connectivity maps: • Maintain network and connectivity where sufficient old forest and protected areas provide X% of channelized and intensified flow. • In areas of deficient connectivity, develop restoration and recruitment strategies and forecast connectivity values to meet threshold estimates at 50, 100, and 200 timescales while accounting for harvest and disturbance consumption scenarios.
How do we measure success?	
Indicator(s)	• Indicators from HRV and FRV landscape connectivity assessments. • Indicators based on a socially agreed upon shared assessments i.e., proportion of landscape unit that meets a defined criterion
Target(s)	1. Targets based on empirical connectivity indicators from HRV and FRV. 2. Indicators based on shared assessments i.e., 70% of landscape units have intensified and channelized flow and or maintaining XYZ corridors over time.
Schedule 'A'	Produce a map of connectivity via Omniscape for routine monitoring and assessment of targets.
What will be reported on?	
Monitoring Strategy	On an annual basis, Omniscape modelling is replicated and used to assess trajectories of connectivity management.
What is the recommended strategy to achieve the outcome?	
Planning Guideline	• Model connectivity using Omniscape and following principles demonstrated in Phase 3 of this project • Compare Omniscape output against HRV and FRV if available

	• Use a connectivity indicator of old patches for comparison • Devise a strategy at a relevant scale i.e., BEC, Landscape Unit or other, that is agreed upon by planning members and land managers • Use a recover/restore, maintain, or enhance strategy based on whether the current planning unit is within or outside of HRV and FRV • If HRV/FRV is not available then the planning committee should set realistic targets based on cooperation to achieve proportional representation of the different flow categories (e.g., Phase 3, Figure 14 and 15). Ideally, proportional representation of connectivity classes should support the managing for connectivity generally, corridors and networks (Figure 1). • Written objectives should reflect modelling and assessment approaches i.e., Phase 3. This ensures consistent assessment and clarity in how land managers develop operational plans to balance economics with connectivity and biodiversity management.
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Appendix A: Landscape resilience infographic

A Framework for Landscape Resilience

The British Columbia Wildfire Service (BCWS), Ministry of Water, Land and Resource Stewardship (WLRS), and Ministry of Forests (FOR) are collaborating to develop a framework to chart pathways of landscape resilience for dry interior plateau ecosystems. The departure of these landscapes and ecosystems from their historical state has resulted in increased fire impacts via fire-deficit conditions, the erosion of ecosystem services, threats to wildlife, wildlife habitat and biodiversity, and increased vulnerability to climate change. **Here we outline a framework to focus landscape management strategies and actions to promote and restore ecological resilience at the landscape and forest stand scale.**

Components of a resilient landscape:

Ecological resilience can be defined as the ability of an ecosystem to retain fundamental structures, processes, functions, and feedbacks following a disturbance. Here we propose managing for the following components to support system recovery and disturbance resistance:

NRV via Spatial Patch Patterning: Through landscape management actions we can align the spatial mosaic of patch patterns to historical conditions that promote how ecosystems tolerate, resist and recover from disturbances.

Fuel Barriers: **Managed:** Managed fuel barriers by Wildfire Risk Reduction (WRR) at the stand and landscape level may help reduce ignition potential, lessen the severity of impacts, limit spread and escapement, and increase the efficiency of fighting fires; these areas can be identified using fuel type maps and by optimization predictions for fuel break networks.

Natural: Natural fuel barriers, which can help limit the spread and escapement of wildfire, may include (1) deciduous vegetation via higher moisture content in foliage; (2) riverscape elements such as beaver habitats and floodplains, wetlands, and water bodies; (3) watersheds and (4) topographical features such as cliffs, alpine areas, rocky outcrops, and glaciers.

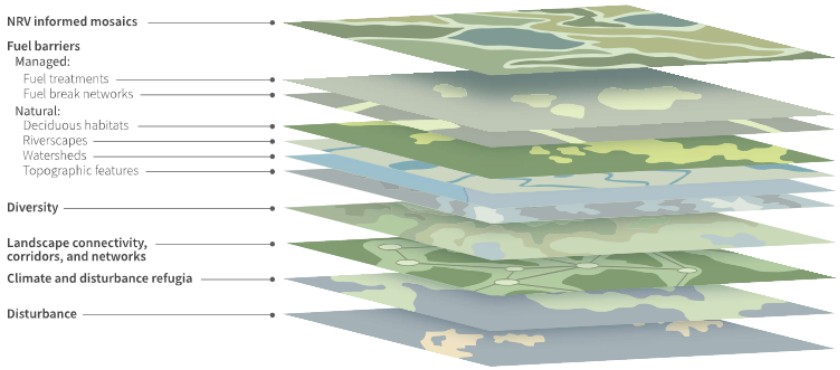
Diversity: Redundancy of patch types and biodiversity hotspots support recovery post disturbance via ecological memory; assisted by species and their interactions that facilitate the restoration of ecological communities.

Landscape Connectivity, Corridors and Networks: Connected landscapes facilitate the movement and exchange of ecological information including genes, species, and nutrients which promote ecosystem recovery post disturbance.

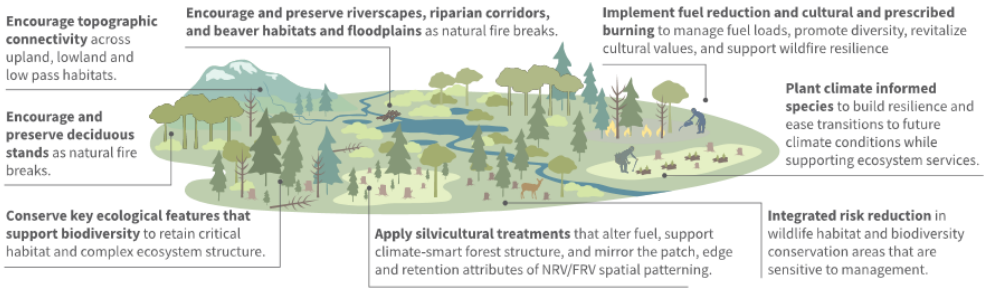
Climate and Disturbance Refugia: Patches that resist climatic alterations and disturbance events are expected to support climate transitions and house remnant ecological identities.

Disturbance: Numerous ecosystems are in a fire-deficit state due to fire suppression activities. Many ecosystems require "good fire" and other disturbances for restoration. Land managers need to encourage fire that promotes resilience via fuel reduction, recruitment of habitat structures, and disturbance-driven biodiversity.

Layered lenses: a holistic approach to landscape management for resilience



Opportunities to support landscape resilience through forest stand scale management:



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BRITISH COLUMBIA | Ministry of Water, Land and Resource Stewardship
BRITISH COLUMBIA | Ministry of Forests